

CALIBRATE THE AMMETER USING POTENTIOMETER

Aim: Calibrate the given ammeter using potentiometer.

Theory:

E.M.F of the Daniel cell, $E = 1.08\text{V}$

Balancing length for Daniel Cell = L in m

Voltage across the standard Resistance, R for current I in the circuit = IR

Balancing length for Voltage across the standard Resistance, $R = l$ in m

$$1.08 \propto L$$

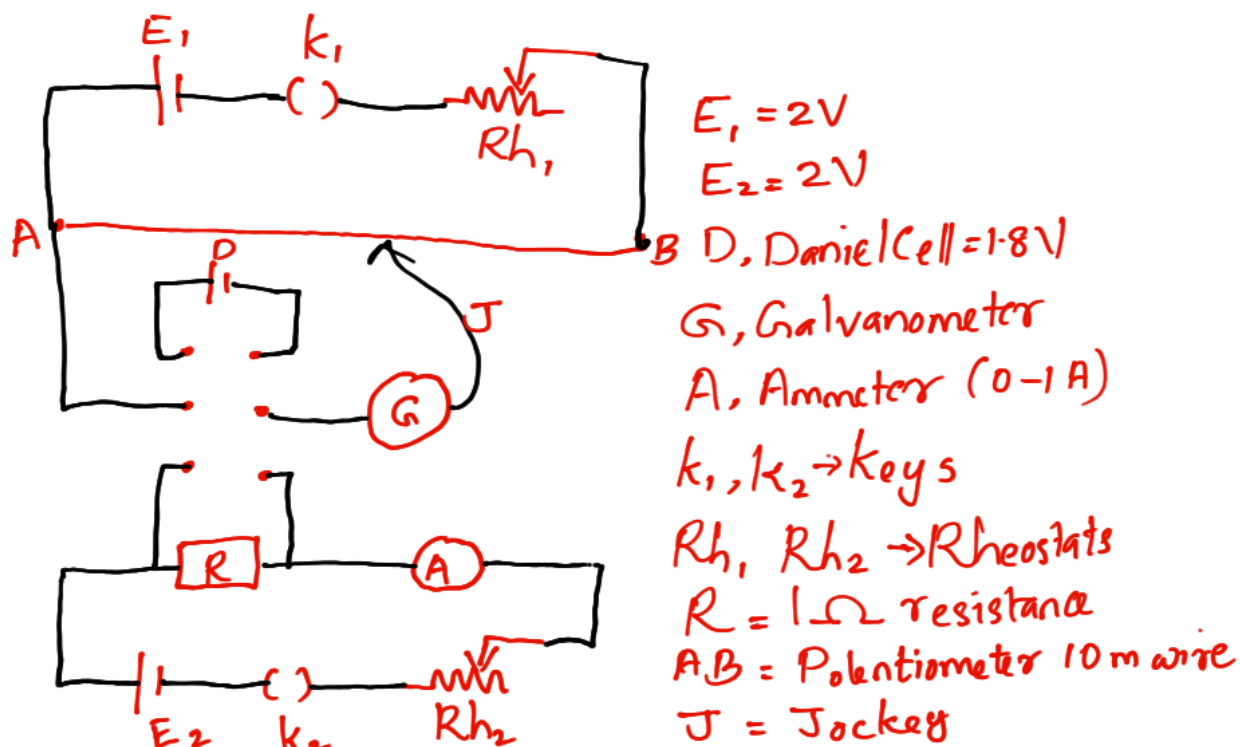
$$IR \propto l$$

$$\frac{IR}{1.08} = \frac{l}{L}$$

Procedure:

- Connections are made as shown in the figure
- Introduce Daniel cell through sixway key and find the balancing length for the known voltage across the Daniel cell. Adjust the rheostat 1 to get maximum balancing length
- Introduce the voltage across the resistance R through sixway key and find the balancing length for 0.1A, 0.2A, 0.8A in the ammeter connected in series with it.
- Calculate the current and compare it with the ammeter reading each time
- Plot the calibration graph with correction $(I - I_0)$ along y axis and I along x axis

Circuit Diagram:



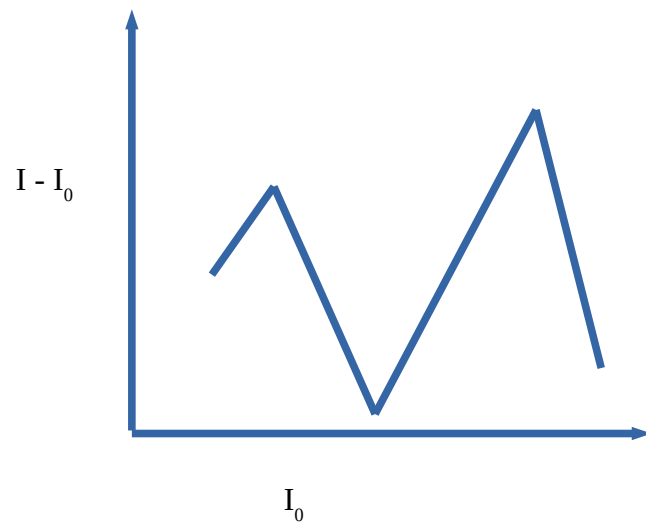
Observation and Tabulations:

E.M.F of the Daniel cell, $E=1.08\text{V}$

Balancing length for Daniel Cell= m

Value of standard Resistance, R= ohm

[illegible]



Result:

The given ammeter is calibrated and the calibration graph is drawn.